Definition of Laplace transform:

$$\mathcal{L}[f(t)] = \int_0^\infty f(t)\, e^{-st}\, dt = F(s)$$

$f(t)$: Input; e^{-st}: Kernel of transformation.

1) find the Laplace transform where

$f(t) = t \ldots 0 < t < 4$ and

$f(t) = 5 \ldots t > 4$ ← Two functions

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t)\, dt$$

$$= \int_0^4 e^{-st} t\, dt + \int_4^\infty e^{-st} 5\, dt$$

Integration by parts:

$$\int u\, dv = uv - \int v\, du$$

we need to find

$u =$

$v =$

$du =$

$dv =$

Pick your u that when we differentiate it gets more simplified. Reduces.

$$\int_0^4 \overbrace{e^{-st}}^{dv}\ \underbrace{t}_{u}\ dt$$

$dv = e^{-st}\,dt$

$u = t$

$du = 1\,dt$

$v = \int e^{-st}\,dt$

$= \boxed{\frac{e^{-st}}{-s}}$

Derivative ::

$\boxed{e^{u(t)} \rightarrow u'(t)\,e}$

$\int e^x\,dx \rightarrow e^x + C$

$\int e^{2x}\,dx \rightarrow \frac{e^{2x}}{2} + C$

$\int e^{-5x} \rightarrow \frac{e^{-5x}}{-5} + C$

$$\frac{e^{-st}}{-s}(t) - \int \frac{e^{-st}}{-s}\,dt$$

$$\left(\frac{t\,e^{-st}}{-s} - \left[\frac{e^{-st}}{s^2}\right]\right)_0^4$$

$$\left(\frac{e^{-st}\,t}{-s} - \frac{e^{-st}}{s^2}\right)_0^4$$

$$\left(\frac{4e^{-4s}}{-s} - \frac{e^{-4s}}{s^2}\right) - \left(-\frac{1}{s^2}\right)$$

$$\frac{4e^{-4s}}{-s} - \frac{e^{-4s}}{s^2} + \frac{1}{s^2}$$

now Solving:

$$\int_4^\infty e^{-st}\, 5\, dt$$

$$= 5 \left| \frac{e^{-st}}{-s} \right|_4^\infty$$

Anything raised to $\infty = 0$

$$\frac{4e^{-4s}}{-s} - \frac{e^{-4s}}{s^2} + \frac{1}{s^2} +$$

$$\left(5\frac{e^{-\infty t}}{-s} + \frac{5e^{-4t}}{+s} \right)$$

$$= \left(0 + \frac{5e^{-4t}}{-s} \right)$$

$$\frac{-4e^{-4s}}{s} - \frac{e^{-4s}}{s^2} + \frac{1}{s^2} + \frac{5e^{-4t}}{-s}$$

final Ans

$$\mathcal{L}[f(t)] = \int_0^\infty f(t)\, e^{-st}\, dt = F(s)$$

Formulas

1) $\mathcal{L}[e^{at}] = \frac{1}{s-a}$ $\quad s > a$ $\quad a$ is a constant

2) $\mathcal{L}[e^{-at}] = \frac{1}{s+a}$

3) $\mathcal{L}[\sin at]$, $\mathcal{L}[\cos at]$

Proving

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{iat} = \cos at + i\sin at$

$\mathcal{L}[\sin at] = \frac{a}{s^2+a^2}$, $\mathcal{L}[\cos at] = \frac{s}{s^2+a^2}$

4) $\mathcal{L}[\sinh at] = \frac{a}{s^2-a^2}$

5) $L[\cosh at] = \frac{s}{s^2 - a^2}$

6) $L[1] = \frac{1}{s}$ $\quad L[2] = \frac{2}{s}$

Q Find laplace transform of $(\sin 3t + e^{-2t})$

$f(t) = \sin 3t + e^{-2t}$

Taking LT on both sides:..

$L[f(t)] = L[\sin 3t + e^{-2t}]$

$L[\sin 3t] + L[e^{-2t}]$

$L[\sin at] = \frac{a}{s^2 + a^2}$ $\quad L[e^{-at}] = \frac{1}{s+a}$

$L[f(t)] = \frac{3}{s^2 + 9} + \left(\frac{1}{s+2}\right)$

$= \boxed{\frac{3}{s^2+9} + \frac{1}{s+2}}$

find the laplace transform of $(\sin 2t - \cos 2t)^2$

$f(t) = \sin^2 2t - 2\sin 2t \cos 2t + \cos^2 2t$

$f(t) = (\sin^2 2t + \cos^2 2t) - 2\sin 2t \cos 2t$

$f(t) =$

$\sin(2\theta) = 2\sin\theta\cos\theta$

$1 - \sin(2t)2$

$1 - \sin(4t)$

$L[f(t)] = L[1 - \sin(4t)]$

$= L[1] - L[\sin(4t)]$

$= \frac{1}{s} - \frac{4}{s^2 + 16}$

Solve L.T of $(\sin t \cdot \cos 2t)$

$f(t) = \sin t \cdot \cos 2t$

$L[f(t)] = L(\sin t) \cdot L(\cos 2t)$ (X)

we can only apply laplace transform on +|−.

$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

find Laplace transform of $(\sin^5 t)$

$f(t) = (\sin t)^5$

(Inverse laplace transform:

$$L[f(t)] = F(s)$$

$$L[t^n] = \frac{n!}{s^{n+1}}$$

$$\boxed{f(t) = L^{-1}[F(s)]}$$

Formula of Inverse laplace transform.

1) $L^{-1}\left[\frac{1}{s}\right] = 1$ $\quad L^{-1}\left(\frac{1}{s^n}\right) = \frac{t^{n-1}}{(n-1)!}$

2) $L^{-1}\left[\frac{1}{s^n}\right] = \frac{t^{n-1}}{(n-1)!}$

3) $L^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$

4) $L^{-1}\left[\frac{s}{s^2-a^2}\right] = \cosh at$

5) $L^{-1}\left[\frac{a}{s^2+a^2}\right] = \sin at$ $\leftarrow$ $a = \sqrt{a^2}$

6) $L^{-1}\left[\frac{a}{s^2-a^2}\right] = \sinh at$

7) $L^{-1}\left[\frac{1}{s+a}\right] = e^{-at}$

8) $L^{-1}\left[\frac{1}{s-a}\right] = e^{at}$

Solve:

find laplace inverse $L^{-1}\left[\frac{3}{s^2} + \frac{2}{s^2+9}\right]$

Taking inverse L.T on both sides

$$L^{-1}[f(s)] = L^{-1}\left(\frac{3}{s^2}\right) + L^{-1}\left(\frac{2}{s^2+9}\right)$$

$$= 3L^{-1}\left(\frac{1}{s^2}\right) + 2L^{-1}\left(\frac{1}{s^2+9}\right)$$

$a^2 = 9$

$a = 3$

$$= 3\left(\frac{t^{2-1}}{(2-1)!} + 2\right) + 2\,\frac{1}{3}\sin 3t.$$

Q Solve

$$L^{-1}\left(\frac{3s+4}{s^2+16} + \frac{2}{s+3} \right)$$

$$\left(\frac{3s}{s^2+16} + \frac{4}{s^2+16} + \frac{2}{s+3} \right)$$

$$L^{-1}\left[\frac{3s}{s^2+16} \right] + L^{-1}\left(\frac{4}{s^2+16} \right] + L^{-1}\left(\frac{2}{s+3} \right]$$

$$= 3L^{-1}\left[\frac{s}{s^2+16} \right] + L^{-1}\left[\frac{4}{s^2+16} \right] + L^{-1}\left[\frac{2}{s+3} \right]$$

$a^2 = 16$, $a = 4$

$$= \boxed{3\cos 4t + \sin 4t + 2e^{-3t}}$$

Using partial fraction $\rightarrow$ $(N < D)$ proper fraction

Type 1 $$\frac{1}{(s+a)(s+b)} = \frac{A}{(s+a)} + \frac{B}{(s+b)}$$

Type 2 $$\frac{1}{(s+a)^2(s+b)} = \frac{A}{s+a} + \frac{B}{(s+a)^2} + \frac{C}{s+b}$$

Type 3 $$\frac{1}{(s+a)(s^2+b)} = \frac{A}{s+a} + \frac{Bs+C}{s^2+b}$$

e.g

$$\frac{2s^2+5s-4}{s^3+s^2-2s} = \frac{2s^2+5s-4}{s(s-1)(s+2)}$$

$$\frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+2}$$

$$A(s-1)(s+2) + B(s)(s+2) + C(s)(s-1) = 2s^2+5s-4$$

Put $s=0$

put $s=-2$

Put $s=1$

$A(0-1)(0+2) + B(0)(0+2) + C(0)(0-1) = 2(0)^2 + 5(0) - 4$

$-2A = -4$

$\boxed{A = 2}$

Put $s = 1$

$A(1-1)(1+2) + B(1)(1+2) + C(1)(1-1) = 2(1)^2 + 5(1) - 4$

$3B = 2 + 5 - 4$

$3B = 3$

$\boxed{B = 1}$

Put $s = -2 \rightarrow \boxed{C = -1}$

$$\frac{2}{s} + \frac{1}{s-1} - \frac{1}{s+2}$$

$$\boxed{L^{-1}\left\{ \frac{2}{s} + \frac{1}{s-1} - \frac{1}{s+2} \right\}}$$

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+2}$$

Denominator

$$A(s+1)(s+2) + B(s+2) + C(s+1)^2 = 4s+5$$

Put $s=-1$

Put $s=-2$

$s+2=0 \Rightarrow s=-2$

$(s+1)^2=0 \quad s+1=0 \Rightarrow s=-1$

$$A(-1+1)(-1+2) + B(-1+2) + C(-1+1)^2 = 4(-1)+5$$

$$B = -4+5 \qquad \boxed{B=1}$$

$$A(-2+1)(-2+2) + B(-2+2) + C(-2+1)^2 = -8+5$$

$$\boxed{C=-3}$$

Now using $s=0$ to find A

$$A(0+1)(0+2) + B(0+2) + C(0+1)^2 = 4(0)+5$$

$$2A + 2B + C = 5$$

$$2A + 2(1) + (-3) = 5$$

$$2A + 2 - 3 = 5$$

$$2A = 6$$

$$\boxed{A=3}$$

$$\frac{3s+1}{(s-1)(s^2+1)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+1}$$

Quadratic

$$A(s^2+1) + (Bs+C)(s-1) = 3s+1$$

Put $s=1$
Put $s=0$

$$A(1^2+1) + (B(1)+C)(1-1) = 3(1)+1$$

$$A = 2$$

$$A(0^2+1) + (Bs+C)(0-1) = 3(0)+1$$

$$A + (B(0)+C)(-1) = 1$$

$$A - C = 1$$

$$2 - C = 1$$

$$2 - 1 = C$$

$$C = 1$$

Quad: 2 new values need 2 no: and another of your choice

Again put $s=2$

$$A(s^2+1) + (Bs+C)(s-1) = 3s+1$$

$$A(4+1) + (B(2)+C)(2-1) = 3(2)+1$$

$$5A + (2B+C)(1) = 7$$

$$5A + 2B + C = 7$$

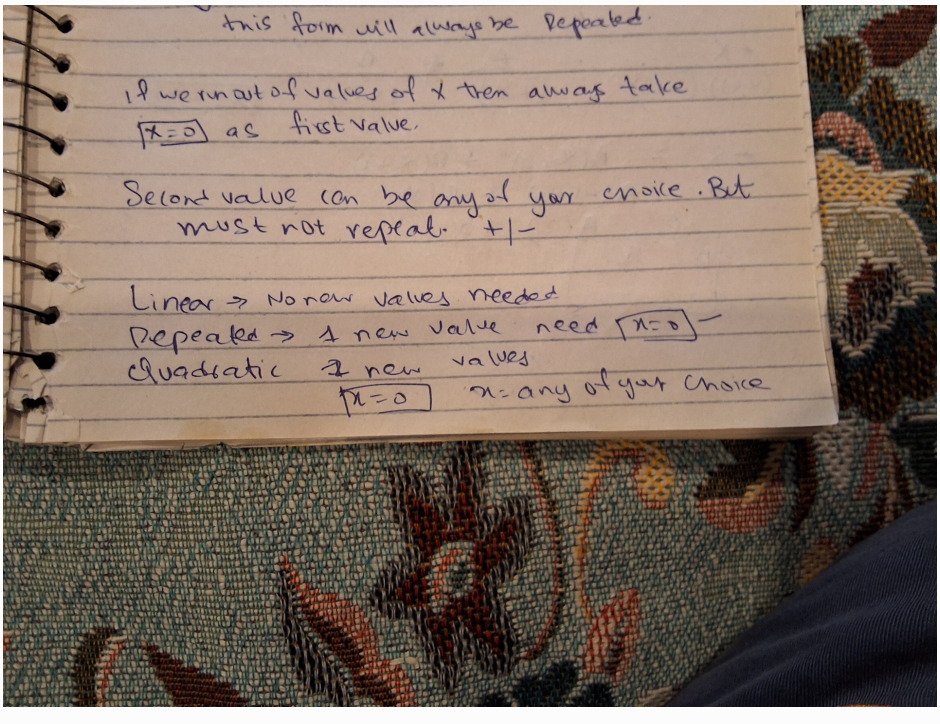
$$5(2) + 2B + 1 = 7$$

$$10 + 2B + 1 = 7$$

$$2B = 7 - 11$$

$$2B = -4$$

$$\boxed{B = -2}$$



Example 4 (7.2)

use laplace transform to solve 1st or 2nd IVP.

$$\frac{dy}{dt} + 3y = 13\sin 2t \qquad y(0) = 6$$

$$L\left[\frac{dy}{dt}\right] + 3L[y] = 13\,L[\sin 2t]$$

$$\boxed{sY(s) - y(0)} \quad \text{where} \quad Y(s) = L[\text{Derivative}]$$

$$sY(s) - 6 + 3L[y] = 13\left[\frac{2}{s^2+4}\right]$$

$$sY(s) - 6 + 3Y(s) = \frac{26}{s^2+4}$$

$$Y(s)(s+3) - 6 = \frac{26}{s^2+4}$$

$$[s+3]\,Y(s) = \frac{26}{s^2+4} + 6$$

$$Y(s) = \frac{26}{s^2+4(s+3)} + \frac{6}{(s+3)}$$

now solve

using partial fractions:

Solve

$$y'' - 3y' + 2y = e^{-4t} \qquad y(0) = 1 \quad y'(0) = 5$$

$$L[y''] - 3L[y'] + 2L[y] = L[e^{-4t}]$$

$$\left[s^2 Y(s) - s\,y(0) - y'(0)\right] - 3\left[sY(s) - y(0)\right] + 2Y(s)$$

$$= \frac{1}{s+4}$$

$$\left[s^2 Y(s) - s - 5\right] - 3\left[sY(s) - 1\right] + 2Y(s)$$

$$= \frac{1}{s+4}$$

$$s^2 Y(s) - s - 5 - 3sY(s) + 3 + 2Y(s) = \frac{1}{s+4}$$

$$Y(s)\left[s^2 - 3s + 2\right] - s - 5 + 3 = \frac{1}{s+4}$$

$$Y(s)\left[s^2-3s+2\right] = \frac{1}{s+4}+s+2$$

$$Y(s) = \frac{1}{s+4(s^2-3s+2)} + \frac{s}{s^2-3s+2} + \frac{2}{s^2-3s+2}$$

Now use partial fractions then $\mathcal{L}^{-1}[\,]$ to solve

(First translation theorem)

To solve $L\left[e^{7t} t^{10}\right]$ → gets very tedious and requires to use Integration by parts.

$$\int_0^\infty e^{-st} f(t)\, dt$$

(Here we use First translation theorem)

$L\left[e^{-2t}\cos 4t\right]$, $L\left[t^{10} e^{7t}\right]$

general equation :: $L\left\{e^{at} f(t)\right\} = F(S-a)$

where $F(S) = L\{f(t)\}$

e.g Find using first value theorem:

$L\left[t^{10} e^{7t}\right\}$ $\qquad L\left[t^{10}\right\} = \frac{10!}{S^{11}}$

now we are going to Subtract a from S.
7 from S.

we always subtract.

$$L\left\{t^{10} e^{7t}\right] = \frac{10!}{(S-7)^{11}}$$

Another e.g

$$\mathcal{L}\left[e^{-2t}\cos 4t\right] = \qquad \mathcal{L}\left[\cos 4t\right] = \frac{s}{s^2+16}$$

$$= \frac{s-(-2)}{(s-(-2))^2+16} \Rightarrow \boxed{\frac{s+2}{(s+2)^2+16}}$$

Inverse form of the theorem:

$$\mathcal{L}\left[e^{at} f(t)\right] = F(s-a)$$

Apply $\mathcal{L}^{-1}$ on both sides.

$$\mathcal{L}^{-1}\left(\mathcal{L}\left[e^{at} f(t)\right]\right) = \mathcal{L}^{-1}\left(F(s-a)\right)$$

$$\boxed{\mathcal{L}^{-1}\left(F(s-a)\right) = e^{at} f(t)}$$

$$\boxed{f(t) = \mathcal{L}^{-1}\left[f(s)\right]}$$

Using inverse form of first translation theorem

Solve

$$L^{-1}\left[\frac{2s+5}{s^2+6s+34}\right]$$

if we cannot middle term break it. use completing square

$$L^{-1}\left[\frac{2s+5}{s^2+6s+(3)^2-(3)^2+34}\right]$$

$$L^{-1}\left[\frac{2s+5}{(s+3)^2-9+34}\right]$$

$$L^{-1}\left[\frac{2s+5}{(s+3)^2+25}\right]$$

Now add 3 to s

$$L^{-1}\left[\frac{2(s+3)+5}{(s+3)^2+25}\right]$$

$$L^{-1}\left[\frac{2(s+3)+5-6}{(s+3)^2+25}\right]$$

Subtracting 6 because we added a $(2s+6)$

$$\mathcal{L}^{-1}\left[\frac{2(s+3)-1}{(s+3)^2+25}\right]$$

$$\mathcal{L}^{-1}\left[\frac{2(s+3)}{(s+3)^2+25}\right] - \mathcal{L}^{-1}\left[\frac{1}{(s+3)^2+25}\right]$$

Fully translated. Every variable s must be added/subtracted with a number.

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now finding $\mathcal{L}^{-1}$

$\mathcal{L}^{-1}$ without shifted

$$\left[\frac{2s}{(s)^2+25}\right] = 2\mathcal{L}^{-1}\left[\frac{s}{s^2+5^2}\right]$$

$$= \boxed{2\cos 5t}$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2+5^2}\right]$$

$$= \frac{1}{5}\mathcal{L}^{-1}\left[\frac{5}{s^2+5^2}\right]$$

$$= \boxed{\frac{1}{5}\sin 5t}$$

$a = -3$ use opposite signs.

a is the amount of shift

$$2e^{-3t}\cos 5t - \frac{1}{5}\sin 5t\, e^{-3t}$$

final Answer

$$\frac{dy}{dt} + 3y = 13\sin 2t \qquad y(0) = 6$$

$$L\left[\frac{dy}{dt}\right] + 3L[y] = 13\,L[\sin 2t]$$

$$\left(sY(s) - y(0)\right) + 3Y(s) = 13\,L[\sin 2t]$$

$$sY(s) - 6 + 3Y(s) = 13\,L[\sin 2t]$$

$$Y(s)\left[s - 6 + 3\right] = 13\,L[\sin 2t]$$

$$Y(s)[s-3] = 13\,L[\sin 2t]$$

$$Y(s)(s-3) = 13\left(\frac{2}{s^2+4}\right)$$

$$L[\sin at] = \frac{a}{s^2+a^2}$$

$$Y(s)[s-3] = \frac{26}{s^2+4}$$

$$Y(s) = \frac{26}{(s^2+4)(s-3)}$$

$$\frac{As+B}{s^2+4} + \frac{C}{s-3} = \frac{26}{(s^2+4)(s-3)}$$

$$(As+B)\cancel{(s-3)} + C(s^2+4) = 26$$

0

$s-3$

$s=0$

$$C(9+4) = 26$$

$$13C = 26$$

$$\boxed{C=2}$$

$s=1$

$$B(-3) + C(4) = 26$$

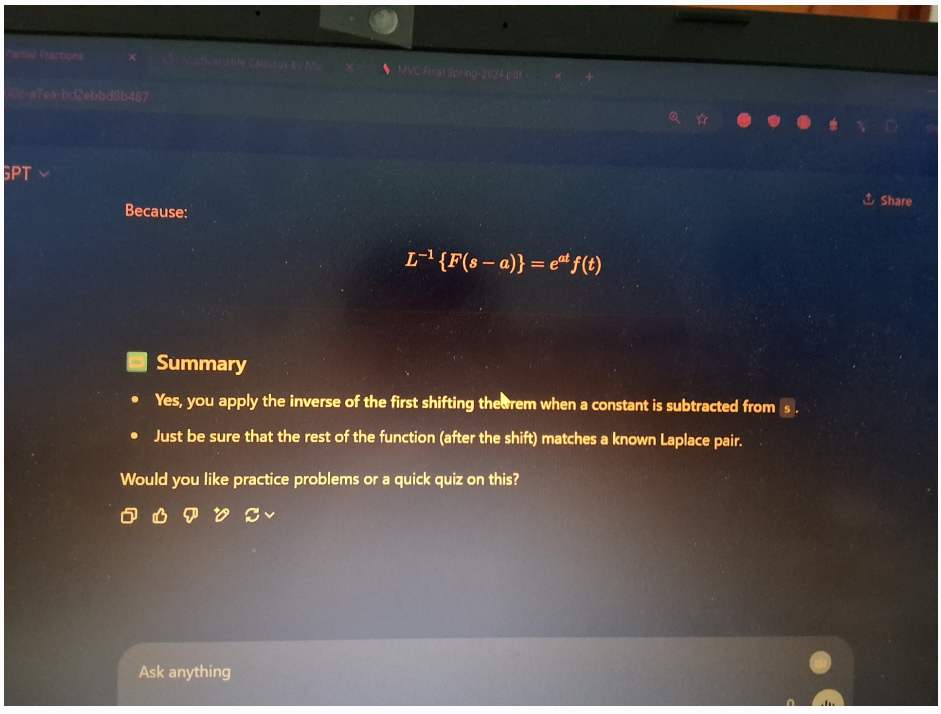
$$B(-3) + 8 = 26$$

$$-3B = 18$$

$$\boxed{B=-6}$$

$$(A-6)(1-3) + 2(5) = 26$$

$$(A-6)(-2) = 16$$



$$-2A + 12 = 16$$

$$-2A = 4$$

$$A = -2$$

$$\frac{-2s - 6}{s^2 + 4} + \frac{2}{s+3}$$

$$\frac{-2(s+3)}{s^2+4} + \frac{2}{s+3}$$

$$-2\mathcal{L}^{-1}\left[\frac{s+3}{s^2+4}\right] + 2\left[\frac{1}{s+3}\right]$$

$$-2\left(\cos(2t)\right) + 2\left(e^{-3t}\right)$$

$$\mathcal{L}[y] = Y(s)$$

$$\mathcal{L}[y'] = sY(s) - y(0)$$

$$\mathcal{L}[y''] = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}[y'''] = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$$